Modeling, Localization, Planning, and Control of an Ackermann Steering Autonomous Vehicle Using ROS2

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***Abstract*—Abstract here**

***Keywords*-Keyword1; Keyword2; ...; KeywordN.**

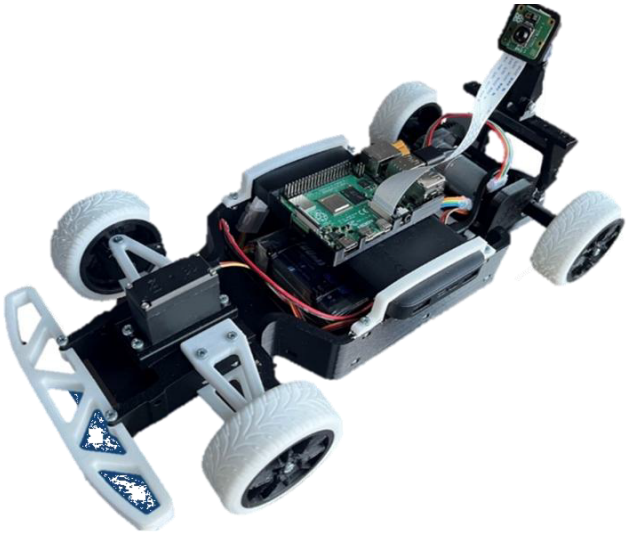

Figure 1: Ackermann Steering Autonomous Vehicle Model Used in Simulation

I. Introduction

Autonomous ground vehicles are a key research area in modern robotics due to their applications in transportation, logistics, and intelligent mobility. They integrate perception, localization, planning, and control to navigate without human intervention, requiring the ability to sense the environment, estimate state, plan safe paths, and execute motion commands.

Among ground vehicle steering mechanisms, the Ackermann configuration is the most widely used in conventional automotive systems. It allows the inner and outer wheels to follow different turning radii during cornering, reducing tire slip and improving maneuverability. Because of its realistic behavior, the Ackermann model is widely used in autonomous driving research and vehicle simulations.

In this work, a scaled autonomous ground vehicle based on the Ackermann steering principle is considered. The vehicle model used in the analysis is shown in Fig. 1. The system is implemented within the Robot Operating System (ROS2) framework, which provides a modular architecture where components communicate through message passing between nodes. Additionally, the Gazebo simulation environment is used to model vehicle dynamics and visualize its interaction with the environment, allowing navigation and control algorithms to be tested safely before real-world deployment.

The autonomous vehicle system analyzed in this report consists of several interconnected modules that together enable autonomous navigation. The localization module estimates the vehicle's position and orientation using sensor measurements. The navigation and planning module generates safe trajectories that guide the vehicle through the environment while avoiding obstacles. Finally, the control module regulates the vehicle velocity and steering angle to accurately follow the desired path.

This report presents a literature review of the key modules required for autonomous Ackermann-steered vehicles. Recent research on localization, path planning, and vehicle control methods is discussed, highlighting how these techniques can be integrated within a ROS-based autonomous driving framework. Understanding the interaction between these modules is essential for designing reliable autonomous ground vehicles capable of operating in complex environments.

II. Literature Review

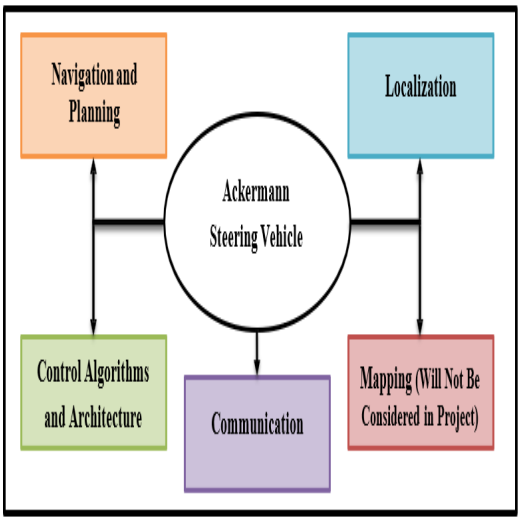

Figure 2: Typical architecture of an autonomous vehicle system.

Autonomous vehicle systems typically use a modular architecture consisting of localization, planning, and control components. These modules enable reliable navigation in dynamic environments. Middleware frameworks such as ROS are commonly used to integrate perception, planning, and control algorithms while allowing different components to communicate through message passing interfaces [1].

A. Localization Methods

Localization is a critical component in autonomous navigation because the vehicle must estimate its position and orientation within the environment. Autonomous vehicles typically rely on multiple sensors such as LiDAR, cameras, GPS, and inertial measurement units (IMUs). Sensor fusion techniques are widely used to combine these measurements and improve localization accuracy. Probabilistic estimation methods such as Extended Kalman Filters (EKF) are commonly used to integrate sensor data and provide robust pose estimation [2]. In addition, LiDAR-based simultaneous localization and mapping (SLAM) algorithms enable vehicles to construct maps while estimating their position [3]. Visual–inertial approaches have also been explored to estimate vehicle motion in GPS-limited environments [4].

B. Navigation and Path Planning

The navigation module is responsible for generating feasible trajectories that guide the vehicle from its current position to a desired destination while avoiding obstacles. Path planning algorithms must consider environmental constraints, vehicle kinematics, and safety requirements.

Several planning strategies have been proposed in the literature for autonomous vehicles. Optimization-based planning approaches compute collision-free trajectories that satisfy vehicle dynamic constraints and minimize travel cost [5]. In addition, sampling-based algorithms and graph-based search methods are commonly used for path generation in complex environments. Many ROS-based navigation frameworks implement real-time planners that continuously update the vehicle trajectory when new obstacles are detected [6]. These algorithms allow autonomous vehicles to adapt their motion in dynamic scenarios while maintaining safe operation.

C. Vehicle Control

The control module is responsible for executing the desired trajectory generated by the planning module while ensuring stable and accurate vehicle motion. Autonomous vehicles require both longitudinal control for regulating vehicle speed and lateral control for maintaining lane position.

Lane keeping control systems are widely studied for autonomous vehicles. Advanced lane keeping controllers use vehicle dynamic models and feedback control strategies to compute steering commands that maintain the vehicle within lane boundaries [7]. Model predictive control (MPC) techniques have also been proposed to improve trajectory tracking performance while respecting vehicle dynamic constraints [8]. Machine learning approaches such as recurrent neural networks have also been explored to improve steering control using sensor data [9].

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